# Eco-Adaptive Home Storage

**Multi-Objective Deep RL for Residential Energy Management**

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CentraleSupélec – Reinforcement Learning Project

## Project Objective

We train a Deep Reinforcement Learning agent to control a home battery over a 24-hour horizon (48 steps of 30 minutes), while balancing three conflicting objectives:

- **Financial optimization**: minimize daily electricity cost through temporal arbitrage.
- **Ecological impact**: reduce $CO_2$ emissions by shifting demand toward cleaner grid periods.
- **Hardware preservation**: limit frequent switching to reduce battery wear.

**Why RL?**

Rule-based policies are simple but brittle under stochastic load, PV production, and dynamic tariffs. DQN learns adaptive action selection directly from interactions and captures non-linear trade-offs between objectives.

## Environment Formulation (MDP)

**State** $s_t \in \mathbb{R}^5$:

$$s_t = [h_t,\ SoC_t,\ P_t^{net},\ C_t^{grid},\ I_t^{CO_2}]$$

**Actions**:

$$a_t \in \{0 : \text{Charge},\ 1 : \text{Discharge},\ 2 : \text{Idle}\}$$

**Reward** (weighted scalarization):

$$R_t = -\left(\alpha \cdot Cost_t + \beta \cdot Emissions_t + \gamma \cdot |a_t - a_{t-1}|\right)$$

**Project specificity**: the $\gamma|a_t - a_{t-1}|$ term explicitly penalizes abrupt action changes, making battery usage smoother and closer to operational constraints.

## Synthetic Inputs

Each episode generates realistic daily profiles:

- household demand with morning/evening peaks,
- PV-like production shape,
- dynamic price and carbon intensity signals.

This procedural generation improves policy robustness by preventing overfitting to a single day pattern.

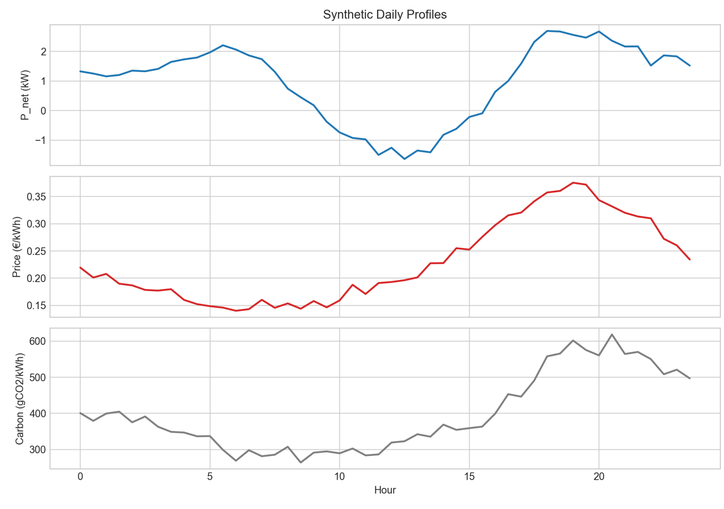


**Figure 1:** One-day synthetic profiles

## DQN Implementation

**Neural model**: MLP $5 \to 64 \to 64 \to 3$, ReLU activations. **Replay buffer**: 20,000 transitions to decorrelate samples. **Batch size**: 64. **Discount factor**: $\gamma = 0.99$. **Target network**: updated every 100 training steps for stability. **Exploration**: $\epsilon$-greedy decay from $1.0$ to $0.05$.

**Bellman Target**

$$Q(s,a) \leftarrow r + \gamma \max_{a'} Q_{\text{target}}(s', a')$$

**Why these choices?**

- **Replay** stabilizes updates by breaking temporal correlation.
- **Target network** reduces oscillations and divergence.
- **Epsilon decay** balances exploration first, exploitation later.

## Training Dynamics

Reward trends upward while TD-loss decreases, indicating progressive learning of value estimates.

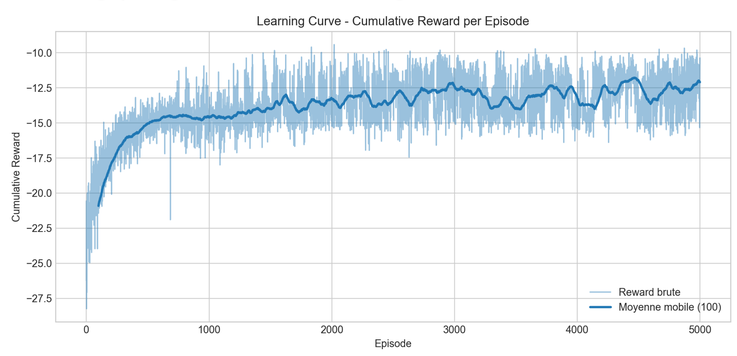


**Figure 2:** Learning curve (reward per episode)

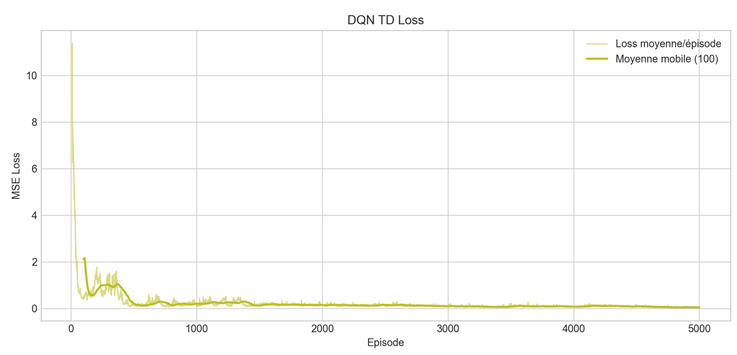


**Figure 3:** Temporal-Difference loss trend

## Policy Analysis

The learned policy exhibits interpretable behavior:

1. **Charge** during low-price or low-carbon windows.
2. **Discharge** during expensive/high-carbon periods.
3. **Avoid excessive switching** due to wear penalty.

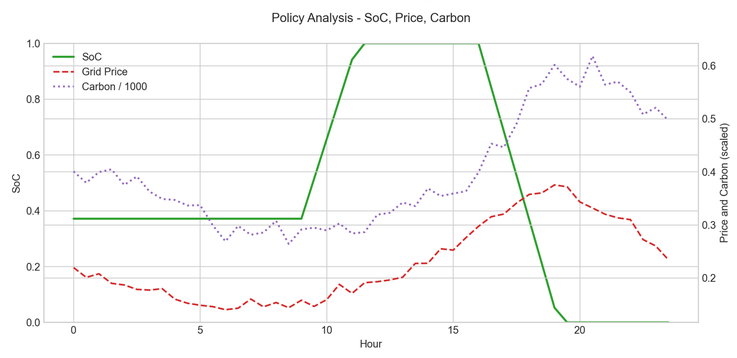


**Figure 4:** SoC dynamics against price and carbon signals

## Baseline and Evaluation Protocol

**Greedy baseline** uses fixed price quantiles:

- Charge when price is in low quantile ($\leq 30\%$),
- Discharge when price is in high quantile ($\geq 70\%$),
- Idle otherwise.

It is a strong and interpretable heuristic, but it ignores long-term effects, carbon context coupling, and switching smoothness.

**Fair evaluation**:

- DQN and Greedy are tested on identical random seeds,
- metrics are averaged over many episodes,
- both are compared to a no-storage baseline.

## Comparative Results

| Strategy | Savings (€/day) | $CO_2$ avoided (kg/ |
|---|---|---|
| Greedy Baseline | 0.47 | 0.10 |
| **DQN (Ours)** | **1.63** | **2.38** |

**Table 1:** Economic and environmental gains

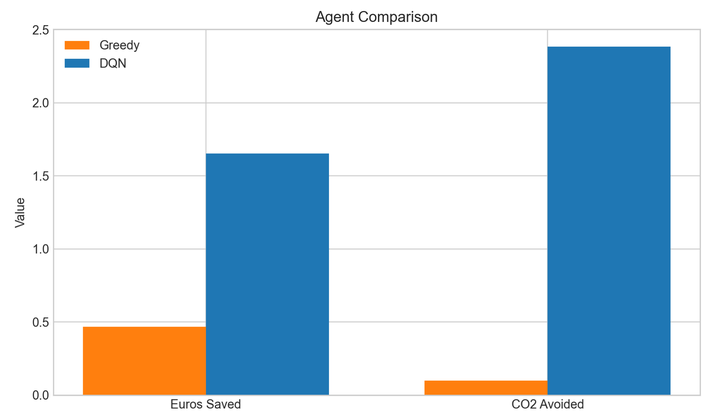


**Figure 5:** KPI comparison: DQN vs Greedy

## Discussion and Limits

- **Adaptability**: DQN reacts better than fixed rules to stochastic profiles.
- **Multi-objective control**: $\alpha, \beta, \gamma$ tune the cost-carbon-wear compromise.
- **Operational realism**: switching penalty discourages micro-cycling.
- **Current limitation**: synthetic data domain

## Conclusion & Next Steps

DQN provides a practical controller for residential battery arbitrage, balancing cost savings, carbon reduction, and battery health in a unified framework.